

## Lecture 16-18

### Introduction to Hypothesis Testing

- we have data  $X_1, \dots, X_n$  generated from a probab. model depending on an unknown param.  $\theta$ .

Null hypothesis  $H_0: \theta \in \Theta_0$

Alternate "  $H_1: \theta \in \Theta_1$

$$\Theta_0 \cap \Theta_1 = \emptyset \quad (\text{empty set})$$

$$\Theta_0 \cup \Theta_1 = \Theta$$

- $H_0 \rightarrow$  no effect, no diff., no improvement, status quo, existing theory, default assumption
- $H_1 \rightarrow$  effect exists, something sed, diff. exists, new thing work or is better/worse
- Reject  $H_0$  or Fail to Reject  $H_0$ .

- Simple hypothesis  $\rightarrow$  where params. are specified

$$H_0: p = \frac{1}{2}, \quad \Theta_0 = \left\{ \frac{1}{2} \right\}$$

- Composite  $\rightarrow$  where several param. values are allowed.

$$H_1: p \neq \frac{1}{2}, \quad \Theta_1 = \left[ 0, \frac{1}{2} \right) \cup \left( \frac{1}{2}, 1 \right]$$

Eg.  $p = P(\text{head}) \quad H_0: p = \frac{1}{2}, \quad H_1: p \neq \frac{1}{2}$

$X = \text{no. of heads}$

$$X \sim \text{Bin} \left( 100, \frac{1}{2} \right)$$

$$E[X] = 100 \times \frac{1}{2} = 50$$

Reject  $H_0$  if  $|X - 50| > c \rightarrow$  how unusual our obs. be before we reject

- errors

reality

decision

result

$H_0$  true

fail to reject  $H_0$

✓

$H_0$  true

reject  $H_0$

Type I error

$H_0$  false

reject  $H_0$

✓

$H_0$  false

fail to reject  $H_0$

Type II error

↗ false alarm  
→ FP

missed  
detection

FN → Type I error = {Reject  $H_0$  /  $H_0$  is ~~false~~ <sup>true</sup>}

FN → Type II error = {fail to reject  $H_0$  /  $H_0$  is false}

eg.  $H_0$ : not pregnant

$H_1$ : pregnant

type I → person is not pregnant, but test says pregnant

type II → person is pregnant, but test says not pregnant

eg. court assumes that the defendant is innocent

$H_0$ : defendant is innocent

$H_1$ : " " guilty

type I → innocent goes to jail

type II → guilty but free

- why we concentrate on rejecting  $H_0$ ? because it is easier to provide strong evidence against a specific default claim than to prove that claim universally.

eg. in airport  $H_0$ : person is harmless

Type 1 → person is harmless, but alarm sounds  
↳ little inconvenience

Type 2 → person is dangerous, but alarm didn't work  
↳ catastrophic issues

eg. 8.9 kg, less  $CO_2$  (claim)

$H_0: \mu = 8.9$ ,  $H_1: \mu < 8.9$

type I → true mean is 8.9, but company says it's lower

type II → true mean is actually lower, we fail to conclude so.

eg.  $p$  = proportion interested

$H_0: p \leq 0.4$

$H_1: p > 0.4$

type II → more people are interested, but they don't consider building it.

- 1 → accepts  $H_0$
- 2 → accepts  $H_1$
- 3 → type 1 →  $H_0$  is true, but reject  $H_0$
- 4 → type 2 →  $H_0$  is false, but fail to reject  $H_0$

$$\alpha = P(\text{type 1 error}) \quad \left| \quad \alpha = P(\text{reject } H_0 \mid H_0 \text{ true}) \right.$$

significance level

$$\beta = P(\text{type II error}) \quad \left| \quad \beta = P(\text{fail to reject } H_0 \mid H_0 \text{ false}) \right.$$

$$H_0: \mu = 500, \quad H_1: \mu \neq 500$$

fail to reject  $H_0$  if  $485 \leq \bar{X} \leq 515$

$[485, 515] \rightarrow$  acceptance / non-rejection region

$\{ \bar{X} < 485 \} \cup \{ \bar{X} > 515 \} \rightarrow$  critical region

eg.  $\sigma = 25, \quad n = 10$

$$H_0: \mu = 500, \quad H_1: \mu \neq 500 \quad \text{fail to reject } H_0$$

$$\alpha = ? \quad 485 \leq \bar{X} \leq 515$$

$$\mu = 500$$

$$\bar{X} < 485 \quad \& \quad \bar{X} > 515$$

$$\alpha = P(\bar{X} < 485 \mid \mu = 500) + P(\bar{X} > 515 \mid \mu = 500)$$

$$SE = \frac{\sigma}{\sqrt{n}} = \frac{25}{\sqrt{10}} = 7.906$$

$$H_0: \bar{X} \sim N(500, 7.906^2)$$

lower limit (485)  $z = \frac{485 - 500}{7.906}$

$$z = -1.90$$

upper limit (515)  $z = \frac{515 - 500}{7.906} = 1.90$

$$\left. \begin{aligned} P(z < -1.90) &= 0.02857 \\ P(z > 1.90) &= 0.02857 \end{aligned} \right\} +$$

$$\alpha = 0.0574$$

eg.  $H_0: p = 0.5$ ,  $X \sim \text{Bin}(100, 0.5)$   $\mu_X = np = 50$   
 $\sigma_X = 5$

$\alpha = 0.05$ ,  $\alpha/2 = 0.025$ ,  $Z_{0.975} = 1.96$

$\frac{|X - 50|}{5} > 1.96 \Rightarrow |X - 50| > 9.8$

$X \leq 40$  or  $X \geq 60$

- $\alpha \rightarrow$  part of  $H_0$  distribu<sup>n</sup> falling in rejection reg.
- $\beta \rightarrow$  part of  $H_1$  distribu<sup>n</sup> " " "non-reject" reg.
- $n \uparrow$ ,  $SE \downarrow$ ,  $\alpha \downarrow$ ,  $\beta \downarrow$

eg. same eg.  $\beta = ?$   $\beta = P(\text{fail to reject} \mid \mu = 520)$   
 $\beta = P(485 \leq \bar{X} \leq 515 \mid 520 = \mu)$

$\mu = 520 \Rightarrow \bar{X} \sim N(520, 7.906^2)$

$Z = \frac{485 - 520}{7.906} \approx -4.43$   $Z = \frac{515 - 520}{7.906} = -0.632$

$\beta = P(-4.43 \leq \bar{X} \leq -0.63)$

$\beta = 0.2643 - 0 = 0.2643$

- $\text{Power} = 1 - \beta$  or  $P(\text{reject } H_0 \mid H_0 \text{ is false})$   
means test successfully detects that the null hypothesis is false.

Power  $\uparrow$ , Type II error  $\downarrow$

high power test is good at finding a real effect.

- $\alpha \downarrow \Rightarrow \beta \uparrow$   
larger reject<sup>n</sup> reg.  $\Rightarrow \alpha \uparrow$ , lower reject<sup>n</sup>  $\Rightarrow \alpha \downarrow$   
 $n \uparrow \Rightarrow \alpha \downarrow$ ,  $\beta \downarrow$   
 $|\mu - \mu_0| \downarrow \Rightarrow \beta \uparrow$

- $p \text{ value} = P(\text{result at least as extreme as obs.} \mid H_0)$   
determined by  $H_1$

-  $p$  value  $\downarrow \rightarrow$  data is difficult to reconcile with  $H_0$ .

$p$  value  $\uparrow \rightarrow$  data is not unusual enough under  $H_0$ .

$p$  value is not  $H_0$  is true. eg.  $p = 0.03$ , it means if we assume  $H_0$  is true, probab. of obtaining a result as extreme as the obs. one is 0.03.

-  $p \leq \alpha \Rightarrow$  reject  $H_0$   
 $p > \alpha \Rightarrow$  fail to reject  $H_0$  } if  $p$  is low,  $H_0$  must go

- upper-tailed test  $\rightarrow$

right tailed when things better  $H_1: \theta > \theta_0$  then unusual obs. are large values.  
 $T = t_{obs}$   
 $p = P_{H_0}(T \geq t_{obs})$

left tailed when things are worse lower-tailed test  $\rightarrow$

$H_1: \theta < \theta_0 \Rightarrow p = P_{H_0}(T \leq t_{obs})$

two-tailed test  $\rightarrow$

when both direc<sup>n</sup> matters  $H_1: \theta \neq \theta_0 \Rightarrow p = 2P(Z \geq |Z_{obs}|)$

equality belongs to  $H_0$

Eg.  $H_0: \mu \leq 1.5$   $H_1: \mu > 1.5$

right-tailed test

critical reg. is in right-tail

Eg.  $H_0: p = 0.6$   $H_1: p \neq 0.6$

two-tailed

critical reg. is divided btw. left & right tails.

- steps -

1. identify param.  $\mu, p, \sigma^2$

2.  $H_0$  &  $H_1$ , equality in  $H_0$

3.  $\mu > \mu_0$  (right),  $\mu < \mu_0$  (left),  $\neq$  two-tailed

4.  $\alpha = 0.05$

5.  $Z = \frac{\bar{X} - \mu_0}{\sigma/\sqrt{n}}$

6.  $|Z| > 1.96$

7. reject  $H_0$  or fail to reject  $H_0$ .